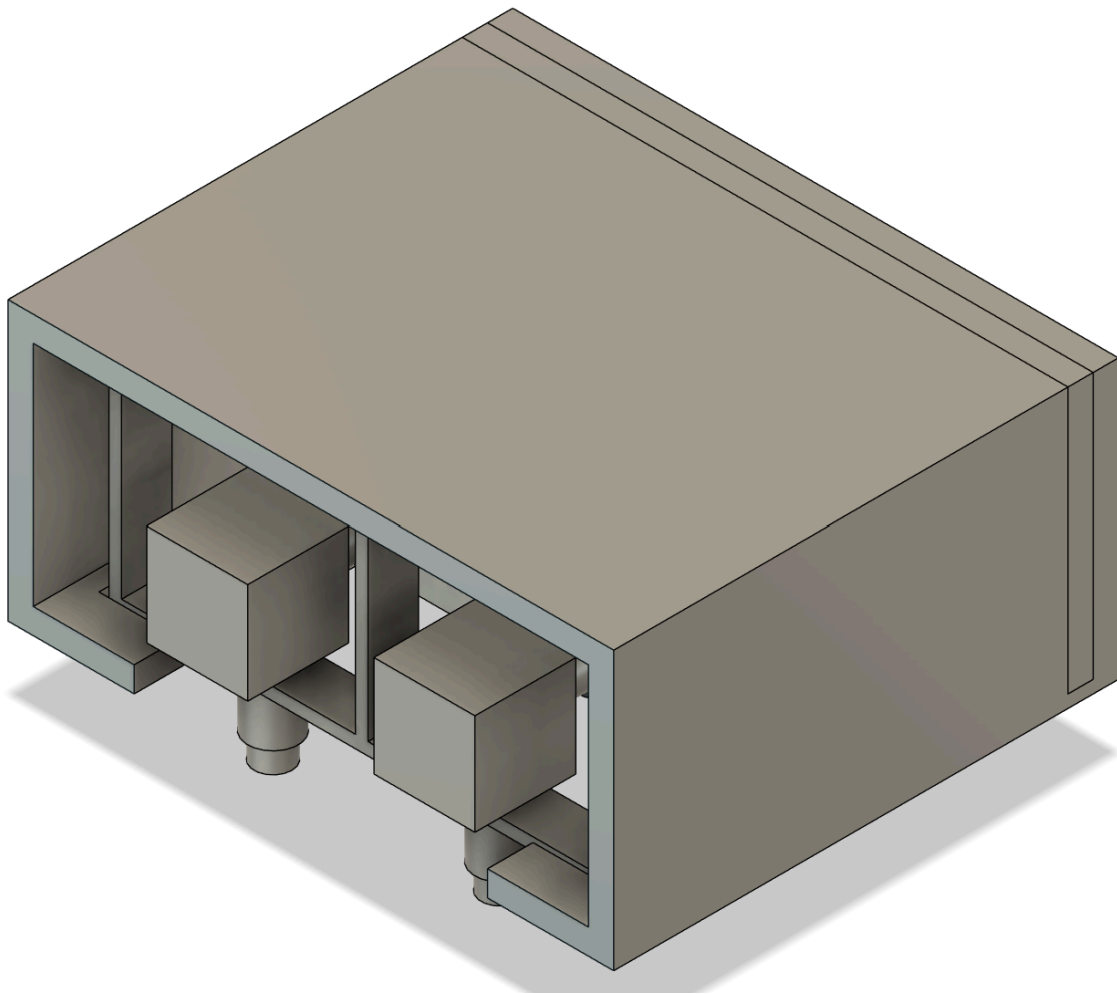


**Design Document for Designing an
Instrument/Calibration setup to Find
Parallel most point
between Two Rotating Cameras**



BY SREERAM PADMESH

Objective

The objective of this assignment is to design a calibration setup to find the parallel most point between two rotating cameras under constraints. The constraints are:

- Distance between the rotating cameras and the surface reference is 150mm
- Angle of view of the lens is 30 degrees
- Tolerance of parallelism alignment is one pixel or 40 micrometers

System

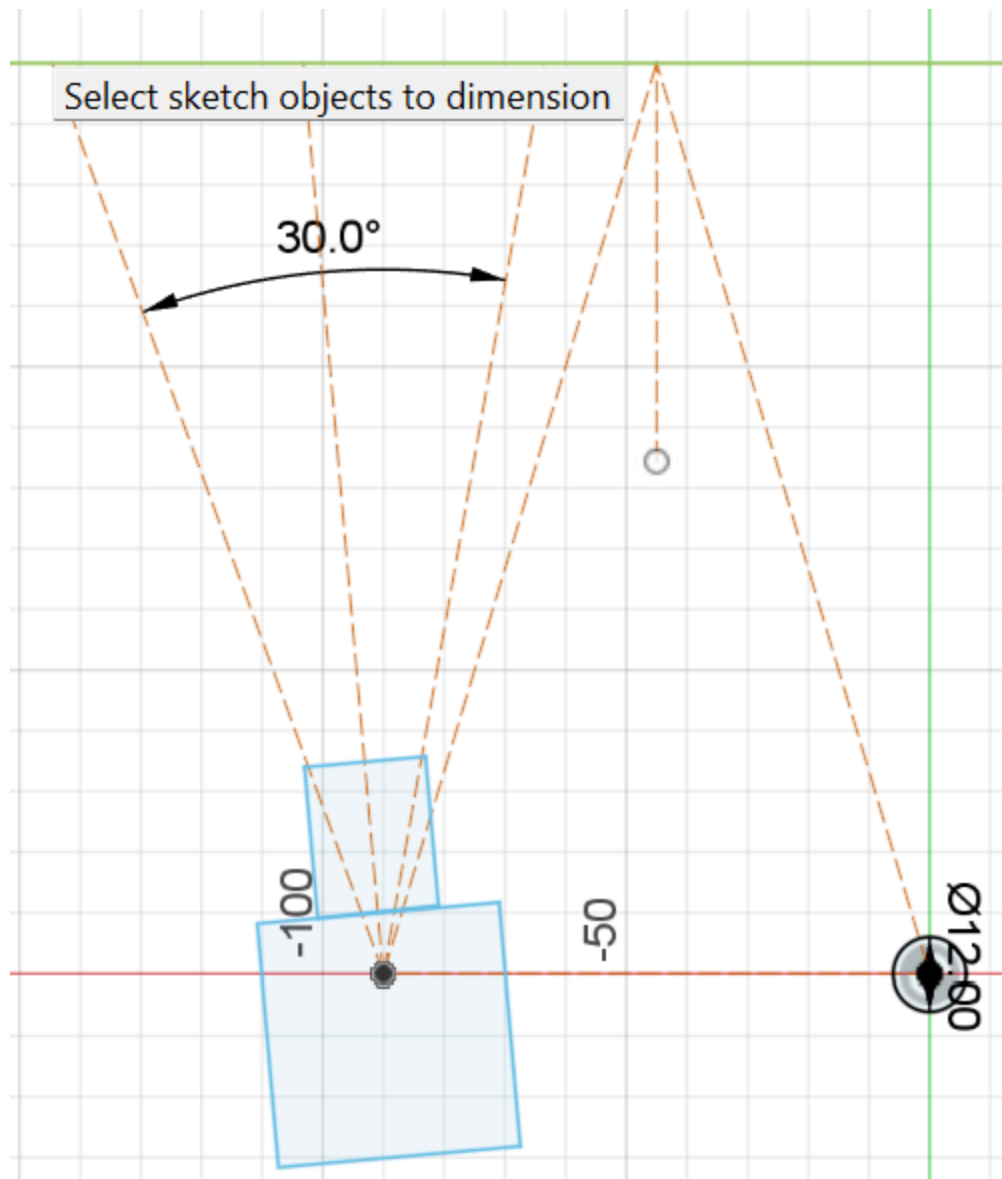
The existing binocular camera assembly features a stacked mechanical design. At the base are the encoders, which sit beneath the motors. A coupler is used to connect the camera and its lens assembly directly to the motor's shaft. Additionally a limit switch is present for both motors to limit the clockwise and anticlockwise movement of the camera.

We must design a calibration system to accurately determine the point of maximum parallelism between the two rotating cameras. To solve this, we will leverage the highly sensitive principles of autocollimation, a metrology concept ideal for measuring small angular differences. An autocollimator is a very expensive tool, but the concepts behind how one works can be translated into our own binocular camera assembly system.

Initial Ideas

Idea 1:

In a dark room, take a light bulb and place it to one side of the camera with a mirror 150mm away. When the camera is rotated away from the light source and mirror, it will not be able to capture the image of the light. Then slowly rotate it one count at a time till the light is visible from the mirror reflection. Knowing that the angle of view of the lens is 30 degrees, we can calculate the rotation required for the camera's optical axis to be exactly perpendicular to the mirror.



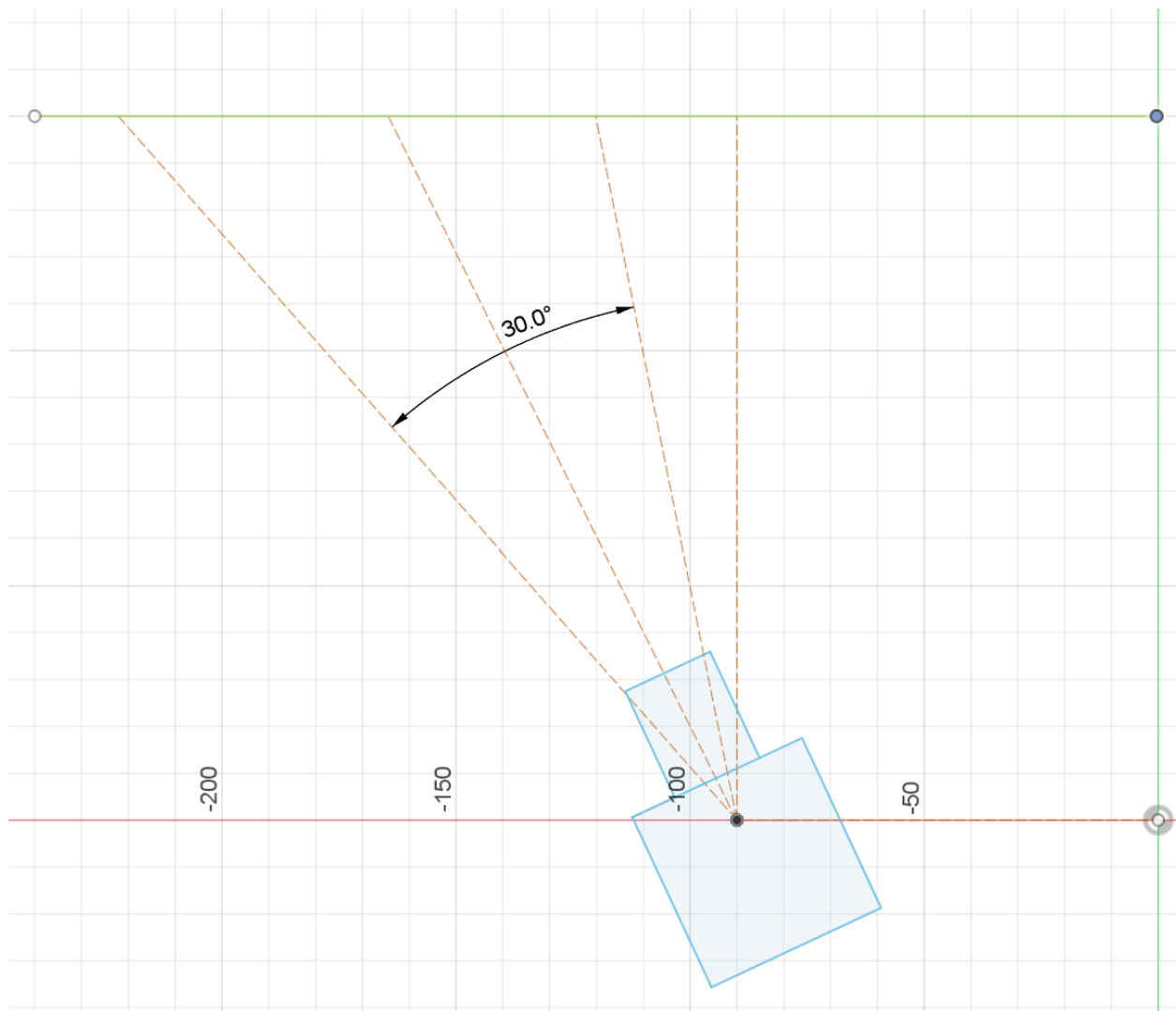
This requires a lot of parameters that needs to be known, angle of view of lens and position of light source with respect to lens placement. We have been given a 30 degree angle of view for the lens and if we can place the light source exactly above the other lens (90mm away). While capturing images for each count, at a certain count the camera will be able to view the mirrored image of the light source, basically when the right line representing the view of lens meets the ray of light that reflects from the mirror to the camera.

After that using maths we find out that the right line representing the view of lens is at an angle of 73.3 degrees with the mirror. We need the angle of the optical axis to be 90 degrees, hence we need the right line to be at 75 degrees angle with respect to the

mirror. So rotating the camera by 1.7 degrees will bring the camera's optical axis perpendicular to the mirror. Do the same for the other camera.

Idea 2:

For each camera, keep a candle light above the camera which acts as a point source. When the camera is facing away, there is no way for the camera to be able to capture the image of the candle light's mirrored image till a certain point while it rotates towards the mirror. This point will be when one of the lines representing the ends of the view of the lens meets the perpendicular with the mirror. After it is pretty simple, rotating the camera and additional 15mm will make the optical axis of the camera perpendicular with the mirror achieving the centre line requirement.



You can see from the above image that the camera cannot view the mirrored image of the point source (candle light) until the right line representing the angle of view of lens

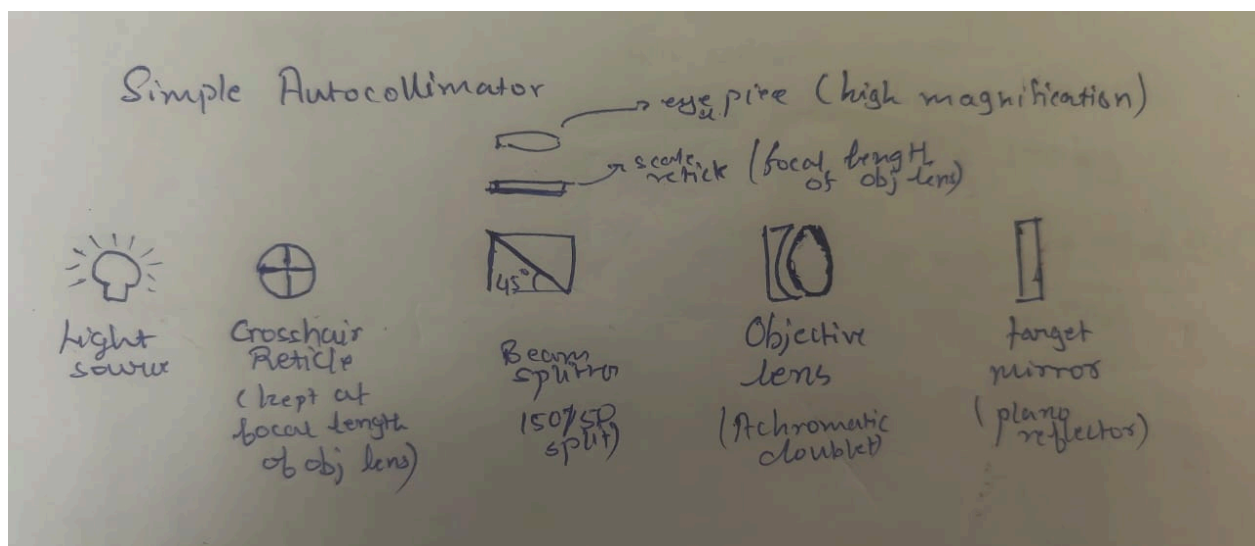
meets the ray of light going exactly perpendicular with the mirror from the candle. After that moving the candle an extra 15 degrees clockwise will allow us to get the centre line and reference it as zero.

But I think the two ideas I explain below which is pretty similar to an autocollimator is much better as there are lesser number of constraints that the calibration setup needs to worry about.

Autocollimation

We must understand the principles of an autocollimator before trying to design a calibration system that utilizes its concepts. Let's look at the most important components of an autocollimator:

- Light Source
- Crosshair reticle
- Beam Splitter
- Objective Lens
- Target Mirror
- Scale Reticle
- Eye Piece



The light source hits the crosshair reticle on its way to the beam splitter which is kept at a 45 degree angle. A beam splitter reflects some of the light towards the scale reticle and eyepiece while allowing the rest of the light to transmit through towards the

objective lens and target mirror. The target mirror reflects back the light towards the beam splitter and onto the eye piece. This allows a person to see how the crosshair reticle is being projected on the mirror and if the mirror is perfectly aligned perpendicular to the autocollimator.

If the crosshair reticle is not directly seen over the scale reticle, and has deviated slightly, then it means that the mirror is at an angle. Slightly rotating the mirror will remove the deviations and this can be done repeatedly till perfect alignment of the mirror is achieved.

This was just a brief overview of what is occurring inside an autocollimator, but there are some important things that need to be taken into consideration before just placing these items as mentioned. Firstly both the crosshair reticle and scale reticle must be kept at the focal length of the objective lens. This is to create collimated light (parallel beams) that the lens will focus back onto the focal plane for viewing. Hence both the reticles are simultaneously perfectly in focus.

Moreover, an autocollimator generally uses an achromatic doublet lens or even ones with more than two lenses attached together which is done to reduce aberrations such as chromatic and spherical aberrations. Chromatic aberration is when certain colours of the light come into focus closer than others while spherical aberration is when rays that start from different distances from the center, focus at different distances from the lens.

About the target mirror, it is better to use a first surface mirror than a second surface mirror as more light gets reflected while also not leaving a ghost image. In a second surface mirror, the light has to pass through a different surface before reaching the mirror surface in which some of the light is already reflected, which creates this dim ghost image. This can be a hindrance while working with extreme precision and hence a first surface mirror is used.

Finally the most important concept of an autocollimator is the formula that the system is built out of:

$$D = 2\theta F \text{ or } \tan(2\theta)F$$

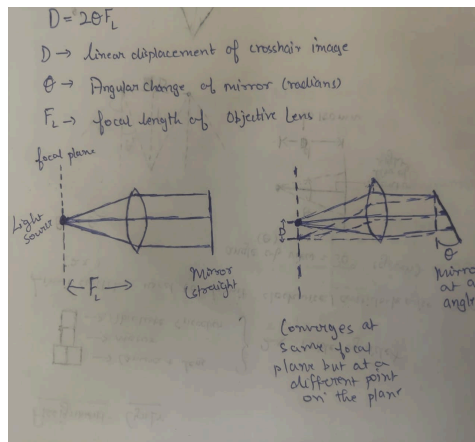
Where:

D - Linear displacement of crosshair image

θ - Angular deviation of mirror in radians (small angles when not using tan)

F - Focal length of objective lens

Using this formula, the mirror angle can be reduced all the way to zero with extreme precision.



Translation of Autocollimation into Our System

Our system has a binocular setup, in which there are cameras, lens, encoders and motors. Our goal is to use the principles of autocollimation and create a calibration setup using the binocular assembly and additional components. Firstly we have to be able to correspond our system with a typical autocollimator. As we have seen an autocollimator consists of the components given below in the left column and the corresponding component in our system is mentioned in the right column.

Autocollimator	Our System
Light Source	Light from Ring Light
Crosshair Reticle	Ring Light
Beam Splitter	Not Required
Objective Lens	Camera Lens
Target Mirror	First Surface Mirror
Scale Reticle	Not Required
Eye Piece	Not Required

Let me explain the system first and I will go in detail about how the corresponding components work with each other. The plan is to mount the camera/binocular setup

onto a flat base and 150mm away place the first surface mirror on the same flat base, at 90 degrees. In front of both the cameras, mount a ring light that faces the mirror. Ideally, the whole setup is closed, with a matte black finish in the walls inside to remove any reflection of light from the wall.

The purpose of this setup is to determine when the optical axis of each camera is exactly perpendicular to the mirror surface. When the ring light is powered, it emits a cone of light toward the mirror. The camera captures both:

1. The direct emission from the ring light (its luminous front surface), and
2. The reflected image of the ring light formed by the mirror.

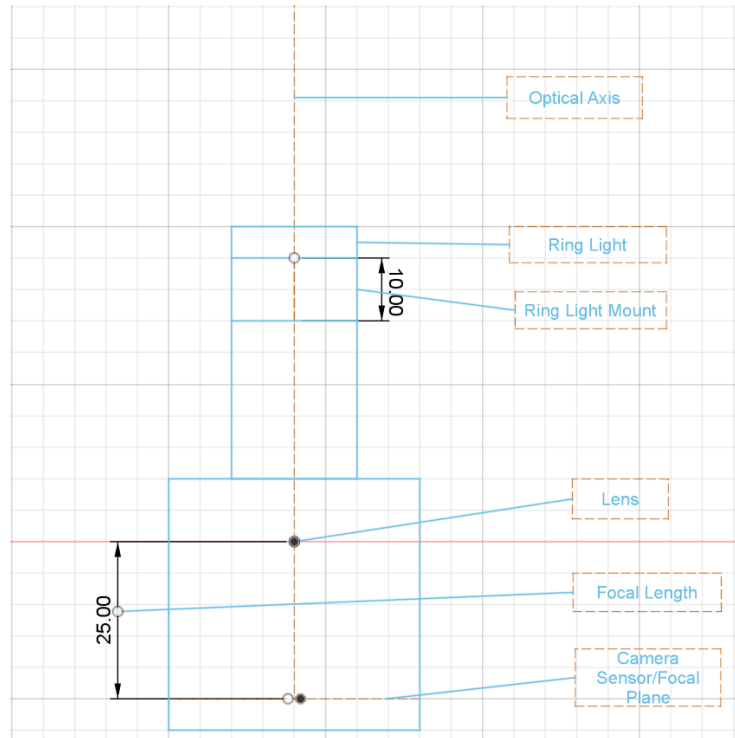
If the camera's optical axis is tilted, the reflected ring appears offset (ellipse shaped) relative to the real ring (circular shaped) in the captured image. When the optical axis is perfectly normal to the mirror, the reflected and direct rings become perfectly concentric. This condition represents the autocollimation null, which defines the camera's centre-line alignment.

Moving ahead on a detailed explanation of the corresponding components. Firstly, the beam splitter, scale reticle and eye piece are not required because unlike a simple autocollimator, there is no requirement for a person to look through an eyepiece and converge the scale reticle with a cross hair reticle, which is visible due to the beam splitter.

The main difference here is using a ring light instead of a crosshair reticle, and this is due to the fact that it creates a bright, uniform pattern creating a visible mirror reflection easy to detect for image processing. The ring light is our reference circle and will be used to check if the mirrored image is exactly concentric with the ring light. When tilted the mirrored image will form an ellipse and will be deviated left or right.

As already discussed while talking about a simple autocollimator, the reason for using first surface mirror remains the same, which is to increase reflected light and remove ghost images.

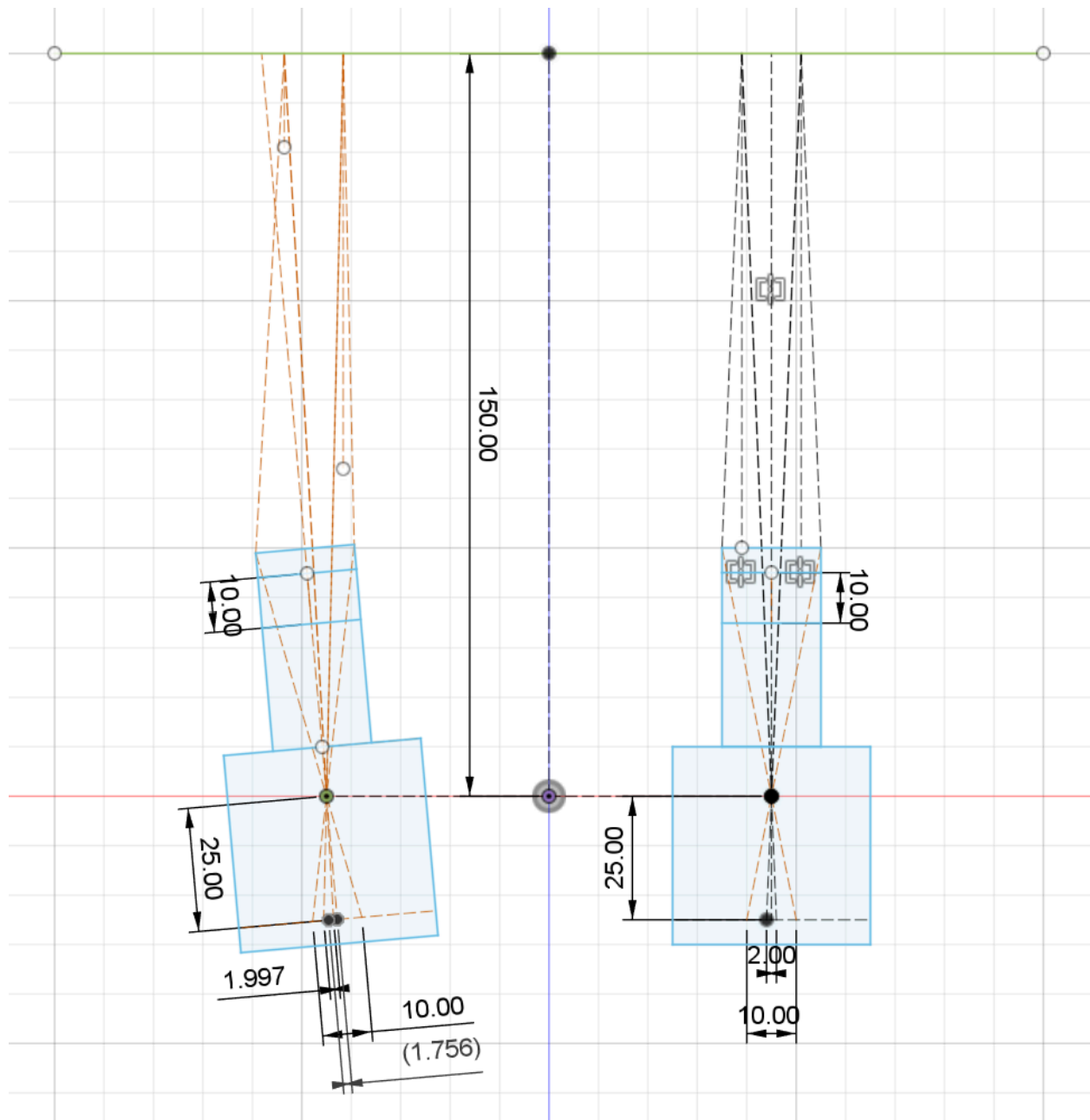
Images to Understand the System



The ring light is kept 10mm away from the lens barrel.

The optical axis and rotational axis are perpendicular to each other and are assumed to meet at a point. The camera sensor/ focal plane is assumed at 25mm behind the lens.

This is just the camera system, the optical illustrations of what the camera will see are on the next page.



On the left you can see a camera that has been tilted by 5 degrees about the rotational axis while on the right you can see the camera when its optical axis is perfectly perpendicular with the mirror.

On the left, we can see that the mirror image 150 mm away is 1.997 mm in length and is slightly deviated towards the left when compared to the 10mm diameter ring light image captured on the camera sensor. While perfectly perpendicular the diameter of the mirrored image is 2 mm but when the camera is tilted an ellipse which is taller than it is longer is created ($2 > 1.997$).

But to make the mirrored image concentric with the ring light we have to go back to the autocollimator's main formula:

$$D = \tan(2\theta)K$$

D - the distance between the 20mm ring circle center and mirrored image ring (ellipse) center on the focal plane

θ - Angular deviation of theta in radians

K - Constant (Not F, because we are not placing our object at focal length from the lens)

The distance can be calculated using image processing of the captured image. The distance calculated by the captured image would be the distance at the focal plane which is required for this formula to work. The constant K can be figured out using ray diagrams using the chief rays and in this case K is close to 10mm (I took the value as 10mm).

$$K = D/\tan(2\theta)$$

$$D = 1.756\text{mm}$$

$$\theta = 5 \text{ degrees}$$

$$K = 1.756/\tan(10) = 9.95877\text{mm} = 10\text{mm}$$

Using these values we can figure out the theta value and how much the camera needs to be rotated to get desired concentric circles.

As an example on the left, when the camera has been rotated 5 degrees to the left, you can see that the centers of the mirrored image and the original ring light captured on the camera sensor is 1.756mm away from each other. K which we have taken to be 10. So theta from the formula can be found to be 4.97980 degrees 360 degrees is 16536 counts, meaning 1 count is about 0.02177 degrees, meaning the camera needs to rotate 228.74 counts = 229 counts.

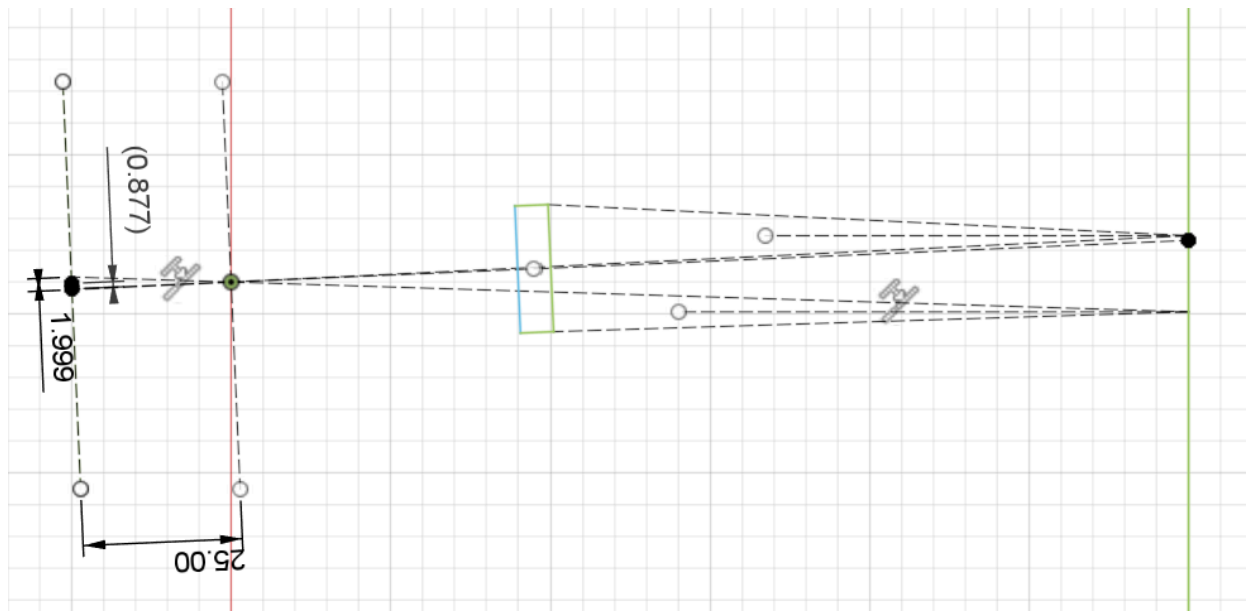
If finding the constant K is not feasible in all scenarios, we can simply rotate the camera count by count slowly till the centers of the ring light and the mirrored image of the ring light are coincident. This is slower because if the camera is deviated by about 5 degrees, we would technically want to take about 230 images for each of the 230 counts to bring it's optical axis perpendicular to the image.

Once the cameras are found to have been centered, read the encoder count and store this value as zero for reference thus later, this can be used to find the angles at which the the cameras converge.

Alternate Idea:

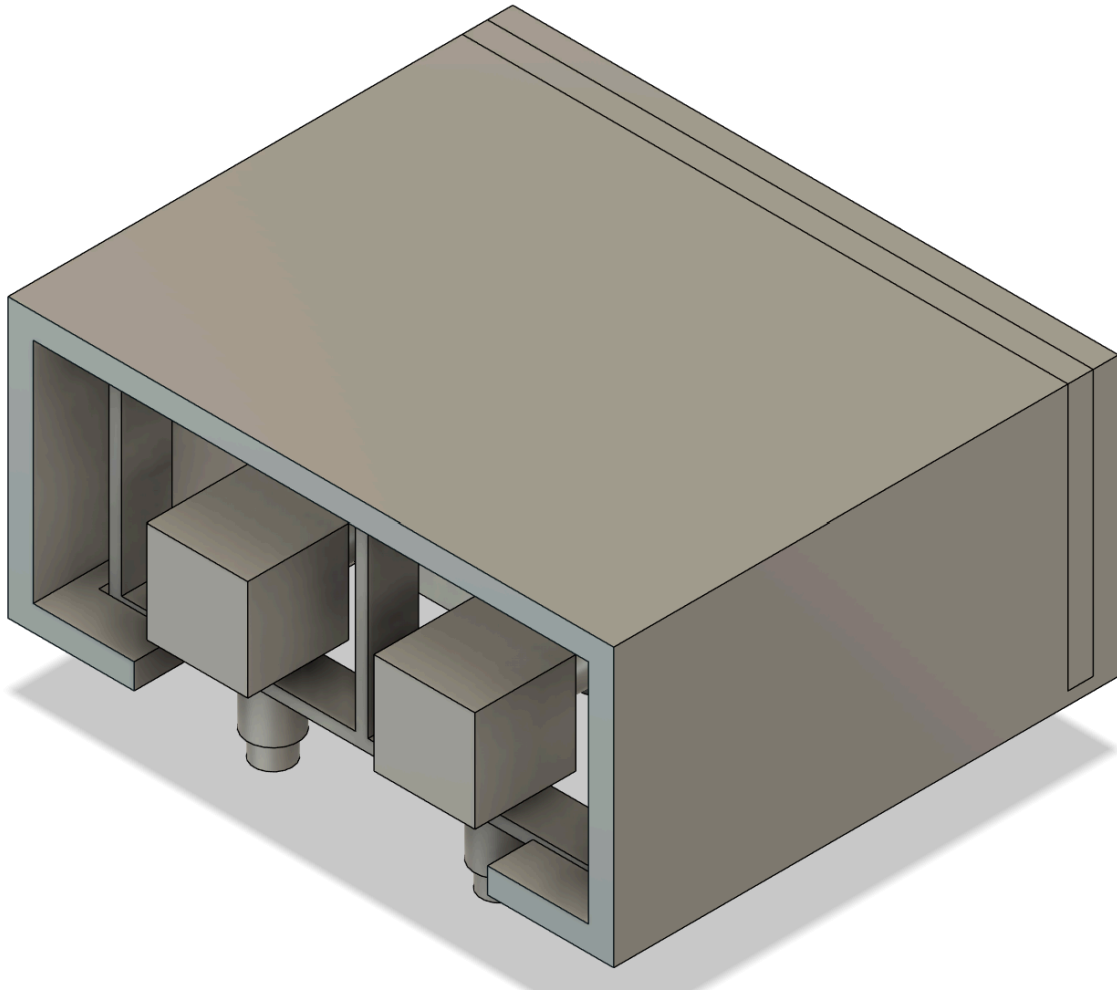
Place a ring light around the camera, and the same way the ring light will be the light source in this setup as well. While going through the previous idea it was clear the deviation/rotation from the mirror makes the mirrored image of the ring light's minor axis distance value greater as the camera rotated away from the mirror. Hence, we can conclude that if the image processor at every count computes the diameter of the reflected ring light, the minor axis length would be at its maximum when the camera's optical axis exactly perpendicular to the mirror.

So I did an additional ray diagram for the same setup at an angle of 2.5 degrees.

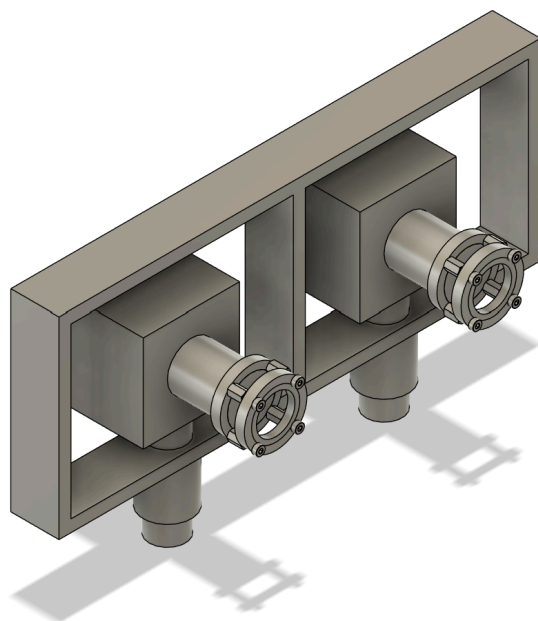
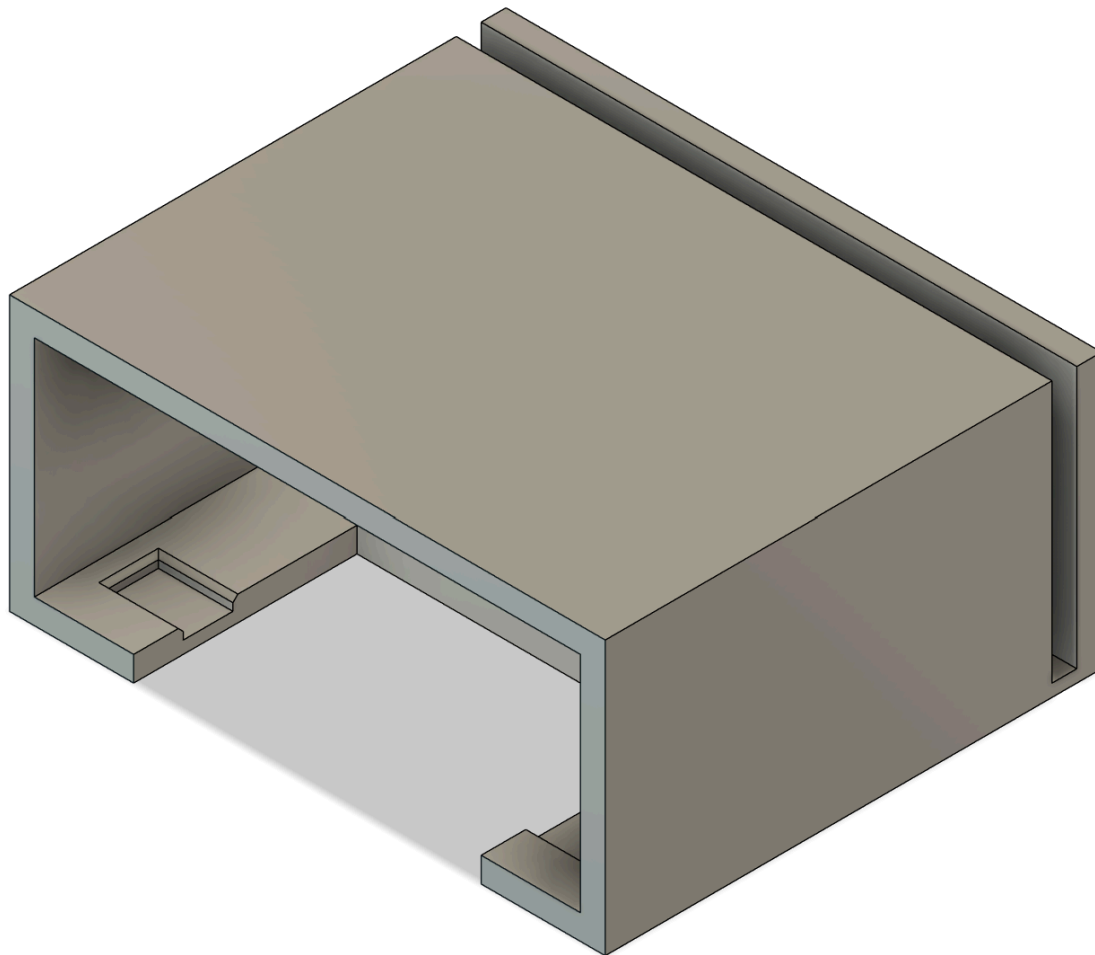


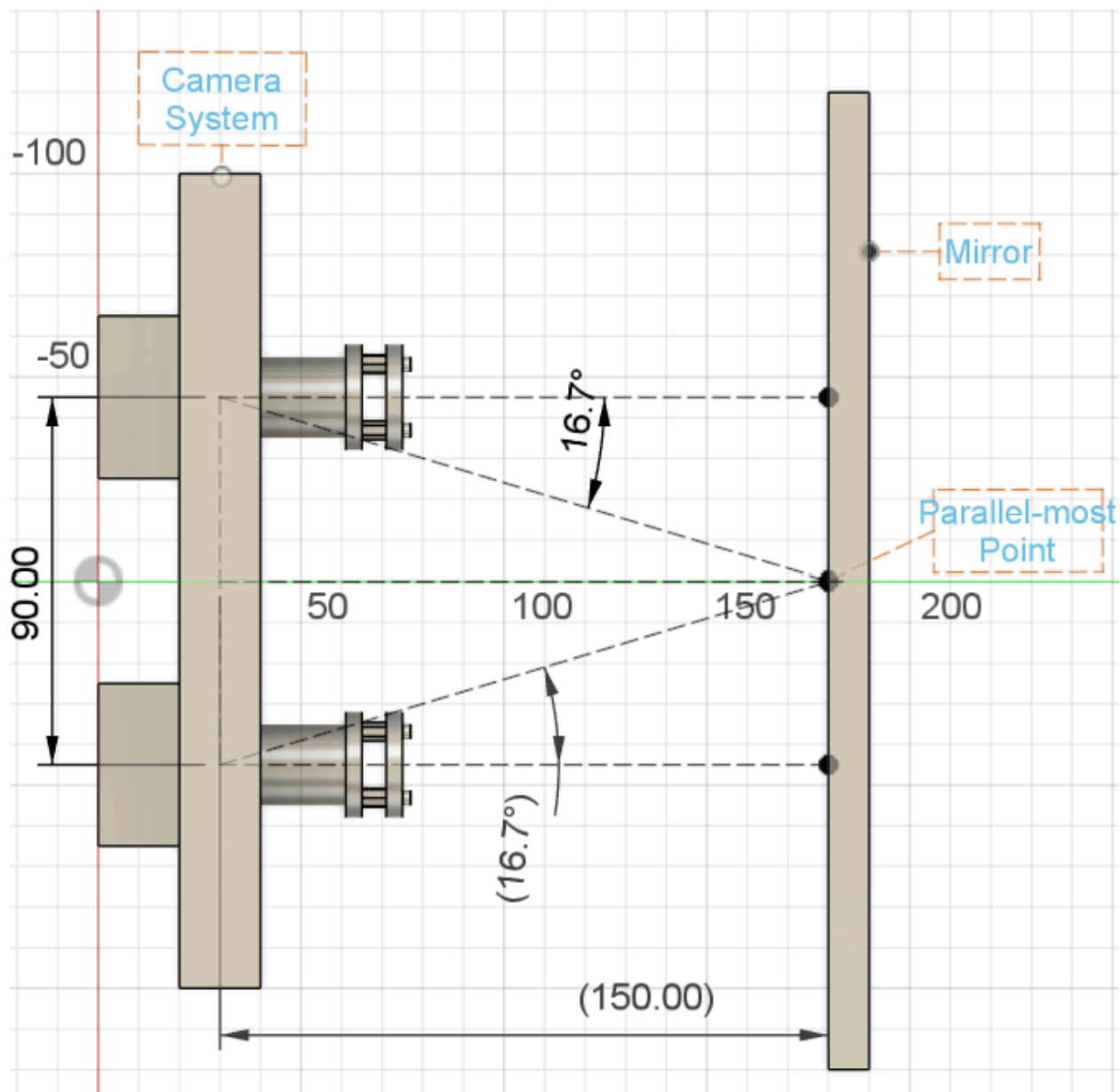
The distance between the edges of the mirrored image, using its chief rays (the rays that pass through the principal point of the lens), increased from 1.997mm when the angle was 5 degrees, to 1.999mm while the angle is 2.5 degrees and finally to 2mm when the angle is 0 degrees. It will decrease from 2mm again when it rotates away from the mirror in the other direction creating a parabolic graph between theta and minor axis distance value.

CAD Assembly

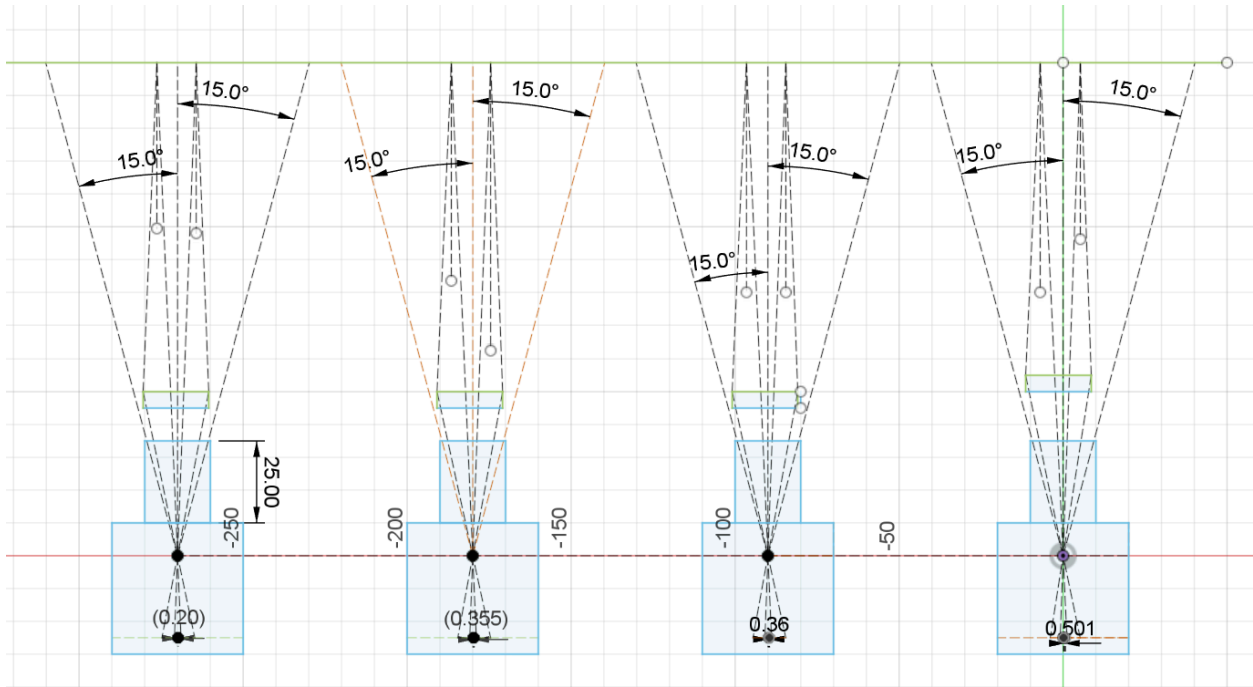


This is the full assembly of the calibration setup, it includes the enclosure, the camera system, the first surface mirror and the ring lights mounted 10mm in front of the camera. The enclosure's inside walls are painted black to reduce reflective behaviour. It has shoulder seats for both the mirror and binocular system of the camera, where they will both sit. The ring light is mounted using 2 annular rings, one fitted onto the barrel of the camera and one kept away using standoffs and a screw. Below you can see the enclosure on its own and the binocular system with the ring light attached to it.





Accuracy and Error Analysis



These are some diagrams representing the rings kept at an offset of 0.5mm, 0.9mm and about 1.4mm (but kept 15mm away). All of these show that when perpendicular with the mirror there is a deviation in the centers of the mirrored image of the ring light with the ring light itself in the captured image. The second diagram and third diagram shows the same 0.9mm offset but with the camera rotated 0.0000153 degrees in the second diagram. Why 0.0000153, using trigonometry, I found out when a mirror is 150mm away, with a maximum of 40 micrometer allowance in the centre line alignment, the allowed angle of deviation is 0.0000153 degrees. This also has shown problems

To conclude for this, I have understood that very small micron level offset of the ring light is allowed, thus choosing the alternate idea is much better for the assignment. This is because, the mirrored image of the ring light will always be largest only when exactly perpendicular with the mirror. But, this makes us think about a different problem as well, what if the ring light itself is slightly tilted across the axis that passes right through the paper/screen, so maybe a combination of both ideas will be required.

Another method that can be used is by finding the principal point of the camera using checkerboard calibration method and using the principal point value as a reference for the mirrored image and getting the center of the mirrored ring light over the principal point.

Mirror tilt should not be much of a problem for attaining cameras that are parallel with each other, but what the mirror tilt may do is that while the cameras may be parallel with each other, both cameras might be rotated clockwise or anticlockwise based on the tilt of the mirror.

Flatness of the base should be within 0.04mm, all points on the base should lie within two planes lying 0.04mm apart so that for upward and downward rotation of the camera, the camera lies in the allotted tolerance range of 0.04mm.

Camera stability will also increase the accuracy of the calibration, so improving stability of camera while the shafts of the motors rotate the camera will be very useful.

Manufacturing and Assembly Tolerances:

Camera seat flatness and mirror seat flatness should be within two surfaces lying 0.04mm away, ± 0.02 mm deviation from nominal to ensure consistent perpendicularity of the optical axis to the mirror plane.

Mirror tilt is allowed but keeping it within a very small angle of about 1-2 degrees should be good as, while the cameras will be parallel with each other, they both will be rotated 1-2 degrees while staying parallel to each other. Basically they will not be parallel to the chassis of the camera system.

Manufacturability and Serviceability

The entire calibration setup is designed for ease of assembly, alignment, and long-term maintenance. All components are based on simple geometries that can be CNC-machined or 3D-printed with high repeatability. The enclosure features shoulder seats to precisely position both the mirror and camera system, ensuring consistent 150 mm spacing during reassembly. The use of detachable mounts allows easy replacement of the mirror or ring light without disturbing the calibrated alignment. Painting the internal surfaces matte black minimizes unwanted reflections and simplifies servicing by making optical artifacts easily visible during inspection. The modular layout also permits quick cleaning of the mirror surface, and lens, supporting repeated calibration cycles without full disassembly.

Conclusion

The most robust calibration strategy for finding the camera's Centre Line (perpendicular to the mirror) is a combined approach that leverages three distinct principles for maximum accuracy and redundancy. We should prioritize the concentric centers method as the primary search criterion: the point of perfect parallelism is achieved when the reflected ring image becomes perfectly concentric with the physical ring light image on the sensor. This null-seeking method is highly sensitive and allows the motor to rotate until the distance D between the centers is zero. This is then validated by the Maximum Minor Axis Method: simultaneously, the image processing algorithm must verify that the minor axis length of the reflected ring (ellipse) is at its maximum value (the diameter of the circle), as this geometric maximum is achieved only when the optical axis is perfectly normal to the mirror. Finally, to ensure the center found is truly aligned with the lens system's fixed reference, the camera's Principal Point (found via initial checkerboard calibration) should be used as an internal reference point to which the center of the reflected ring image must align, minimizing errors from non-ideal ring light placement. This three-point verification maximizes confidence in setting the encoder's zero reference count.

After finding the center using basic trigonometry, we can find if both the cameras rotate by 16.7 degrees clockwise and anticlockwise respectively they will meet at the parallel most point, converging 150mm away. So after finding the reference points for the center line, rotate both cameras 767 counts to converge them at the parallel most point.